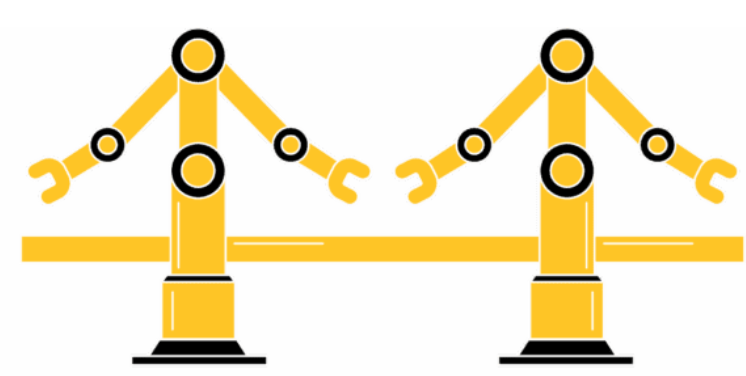




IROS 2026
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PI-EAT-SLAM: Physics-Infomed Edge-Aware Thermal Gaussian Splatting SLAM

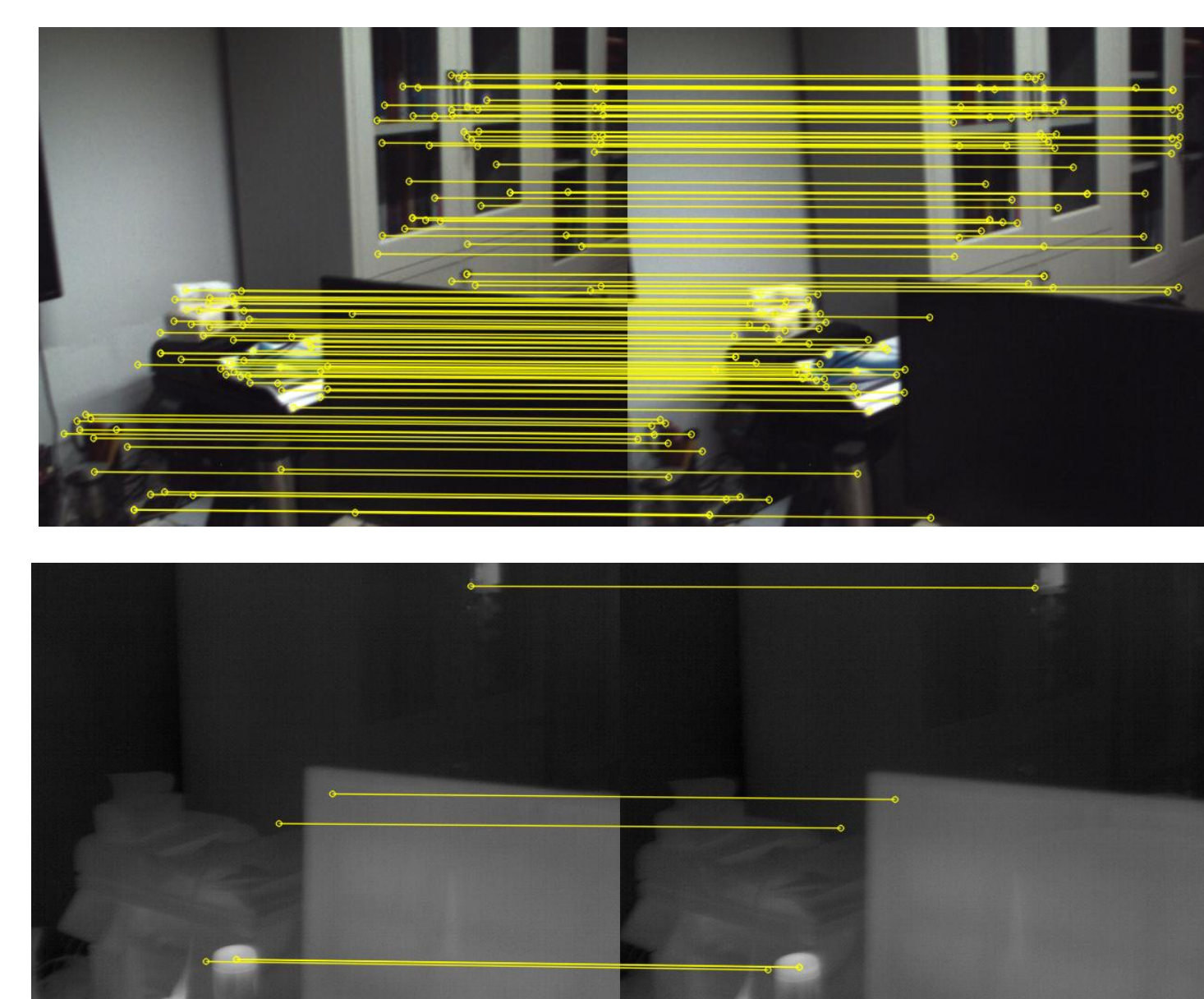
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Hyeonji Shin¹, Changhyeon Kim², Hyeonbeom Lee³, Pyojin Kim^{1*}

¹Gwangju Institute of Science and Technology, ²Holiday Robotics, ³Ajou University

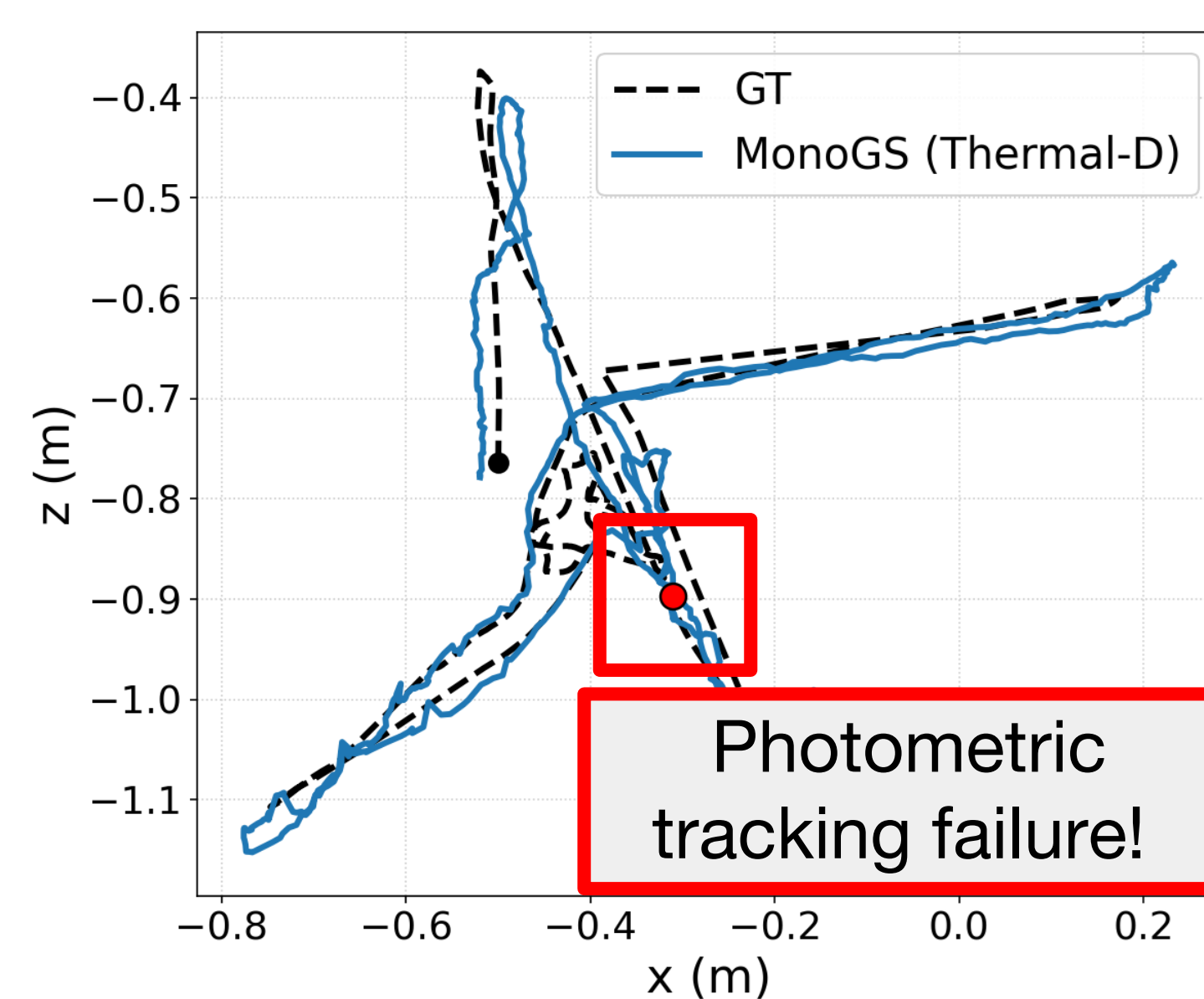


Motivation

- Conventional visual SLAM systems primarily rely on **point feature matching** or **photometric loss**.
- Existing thermal SLAM methods heavily depend on **auxiliary sensors** like IMU or LiDAR.

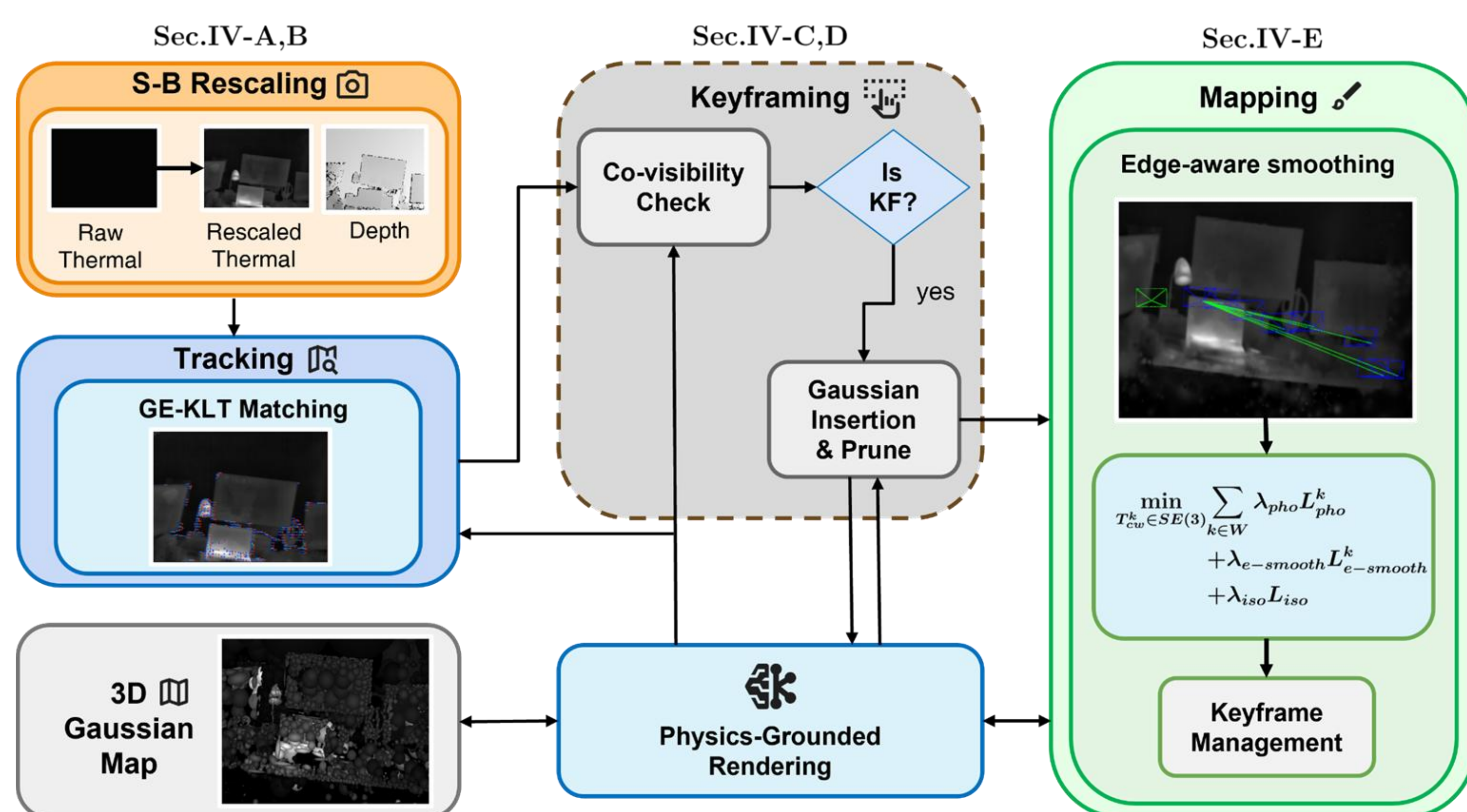


SIFT+RANSAC



MS dataset: desk2-xyz

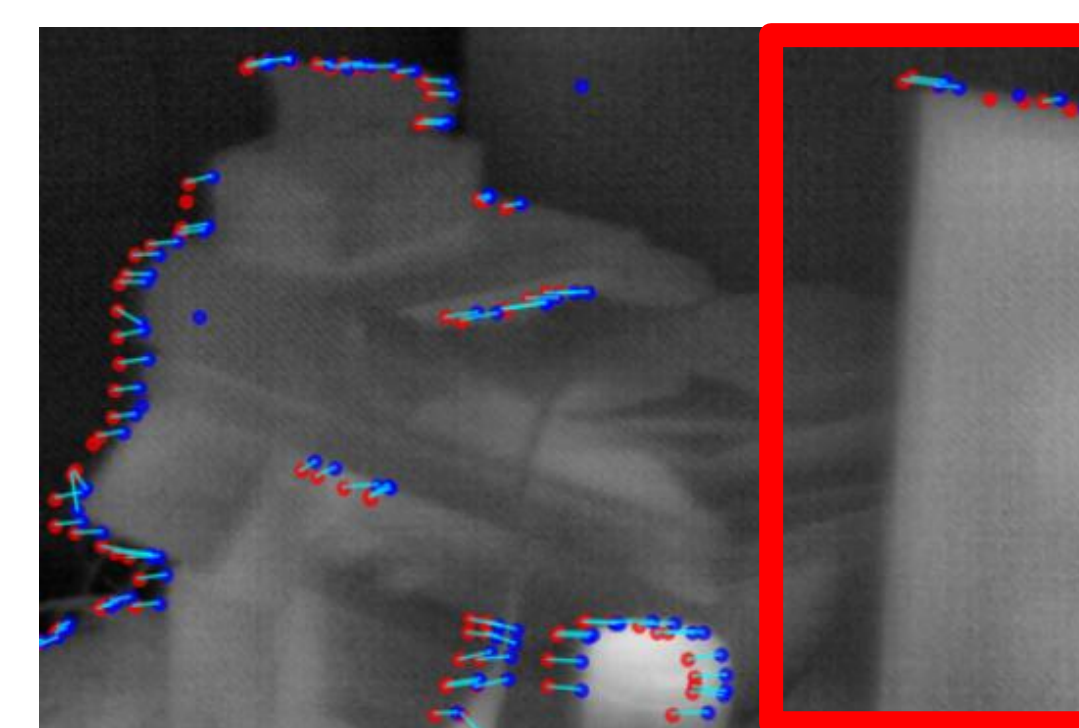
Contributions



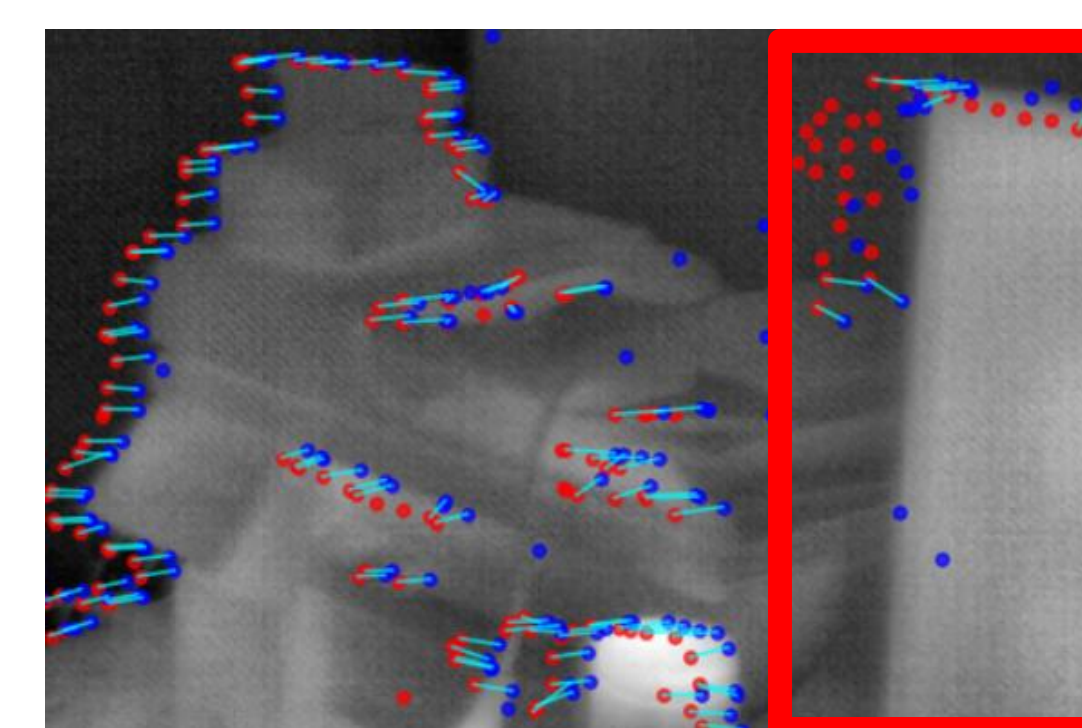
- Stefan-Boltzmann Law-based thermal rescaling
- Gradient-aware Edge-KLT (GE-KLT) loss for tracking
- Edge-aware smoothness loss for mapping

Stefan-Boltzmann rescaling

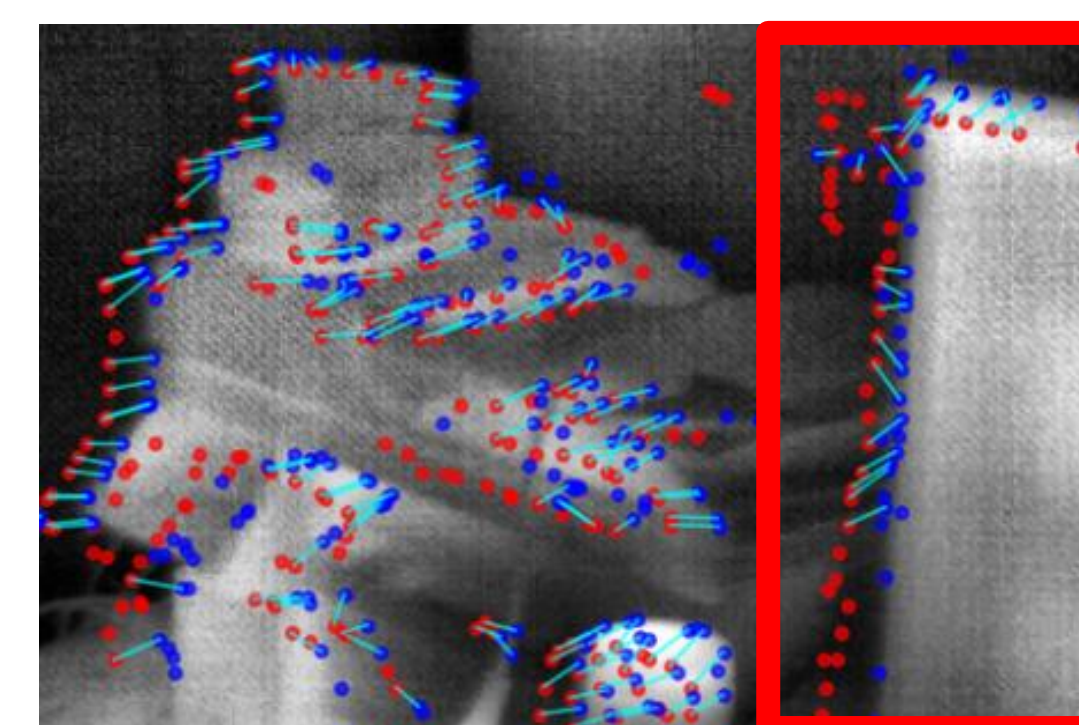
- Stefan-Boltzmann Law: $L(x, y) = \epsilon \sigma T(x, y)^4$



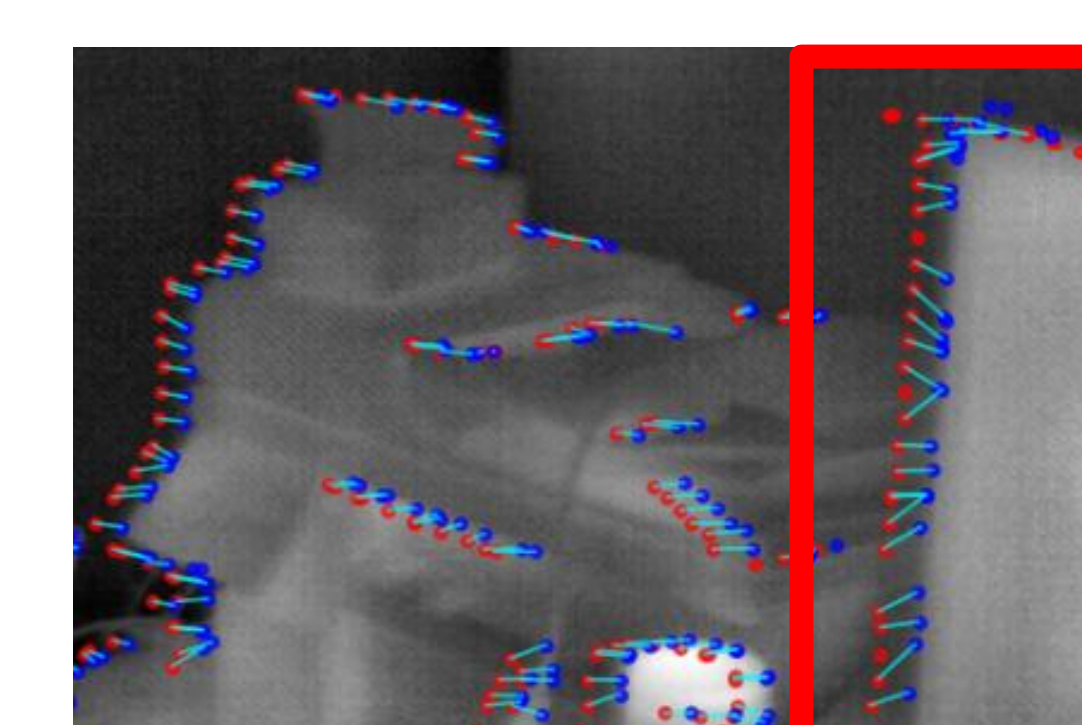
Original



Min-max scaling



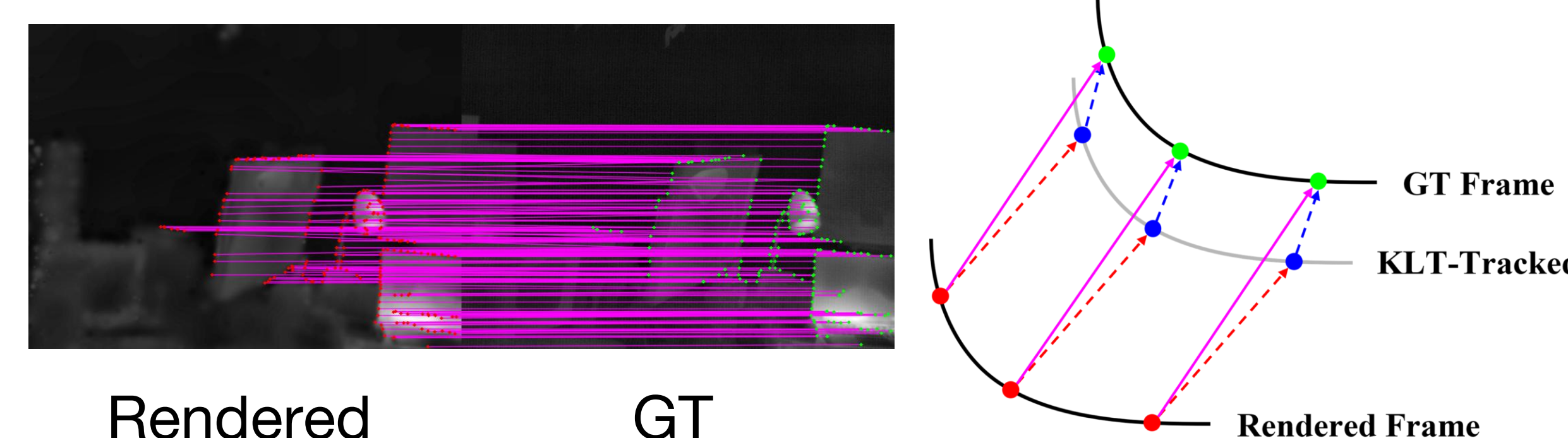
Fieldscale



S-B rescaling

Edge-based optimization

- Tracking:** We propose **GE-KLT loss** which integrates E-KLT and nearest neighbor matching.

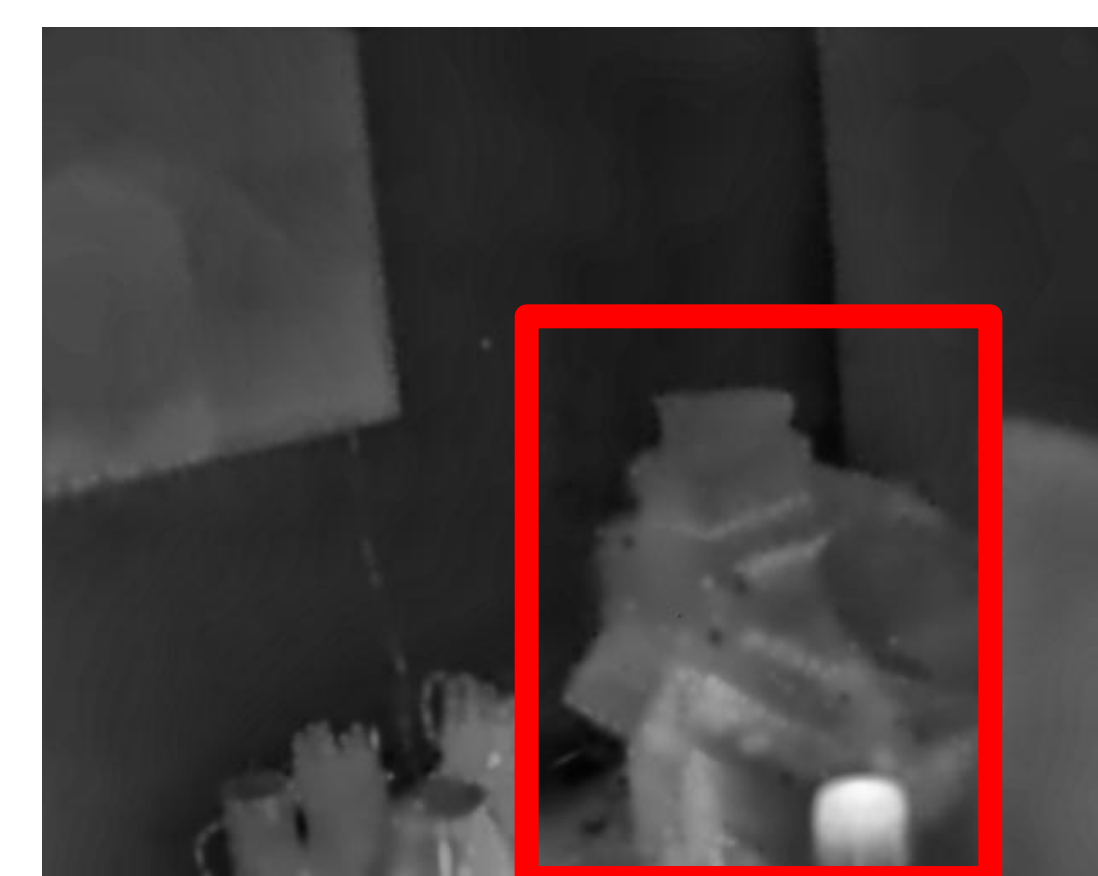


Rendered

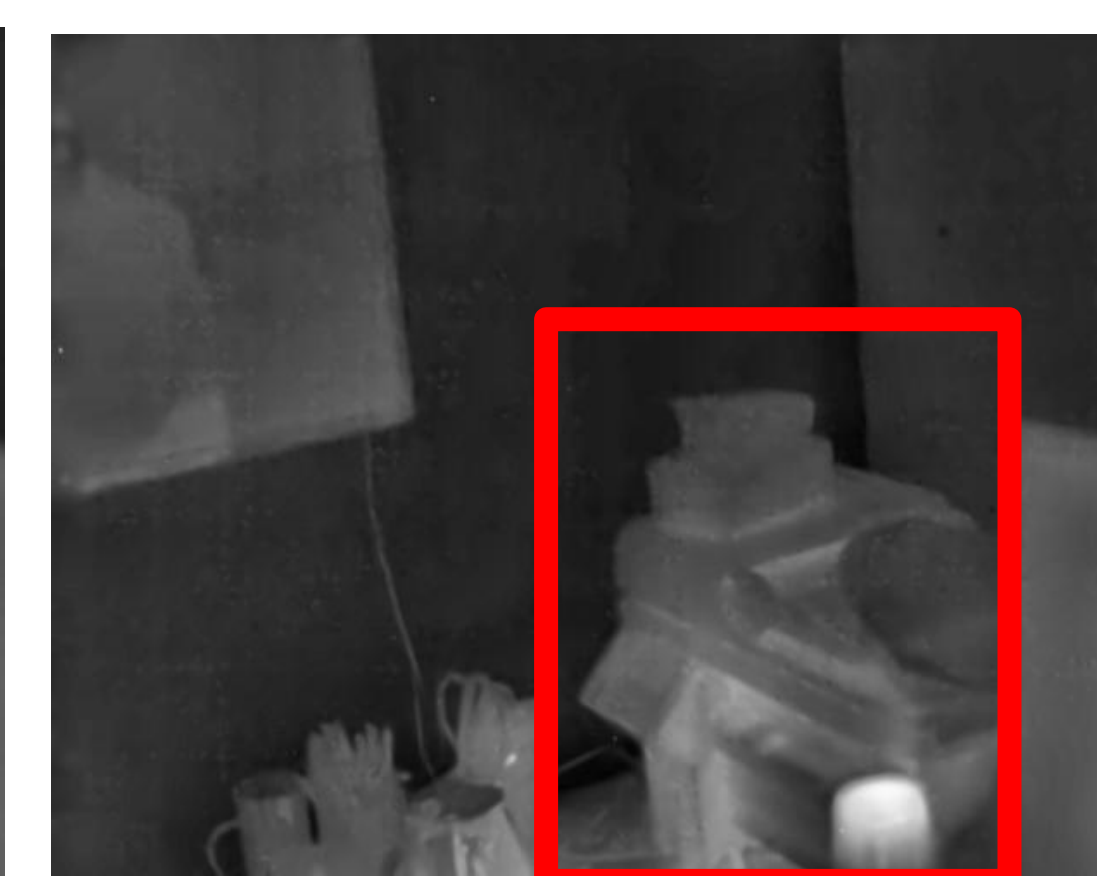
GT

Rendered Frame

- Mapping:** **Edge-aware smoothness loss** selectively smooths non-edge regions.



Smoothness loss

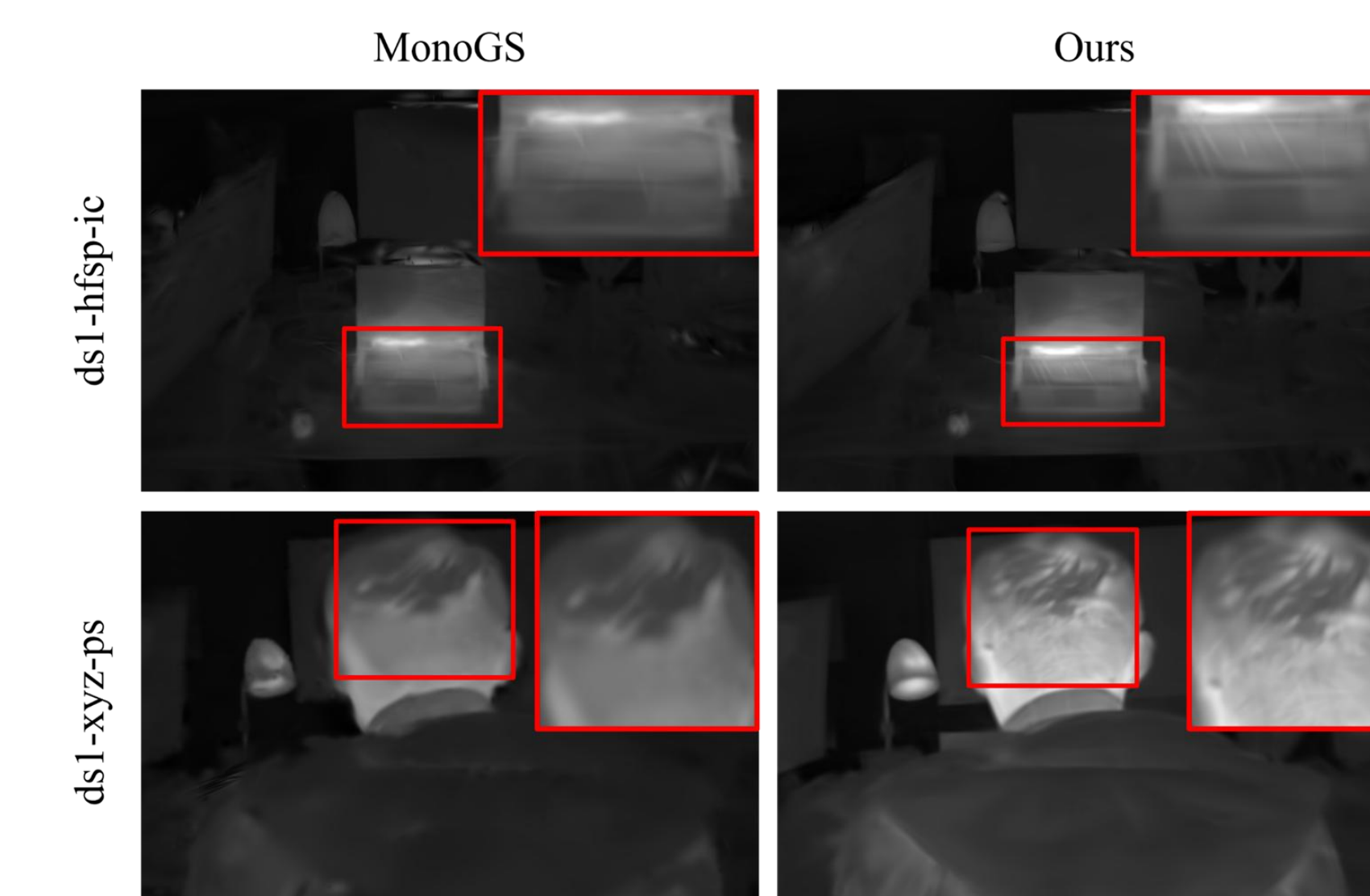


E-smoothness loss

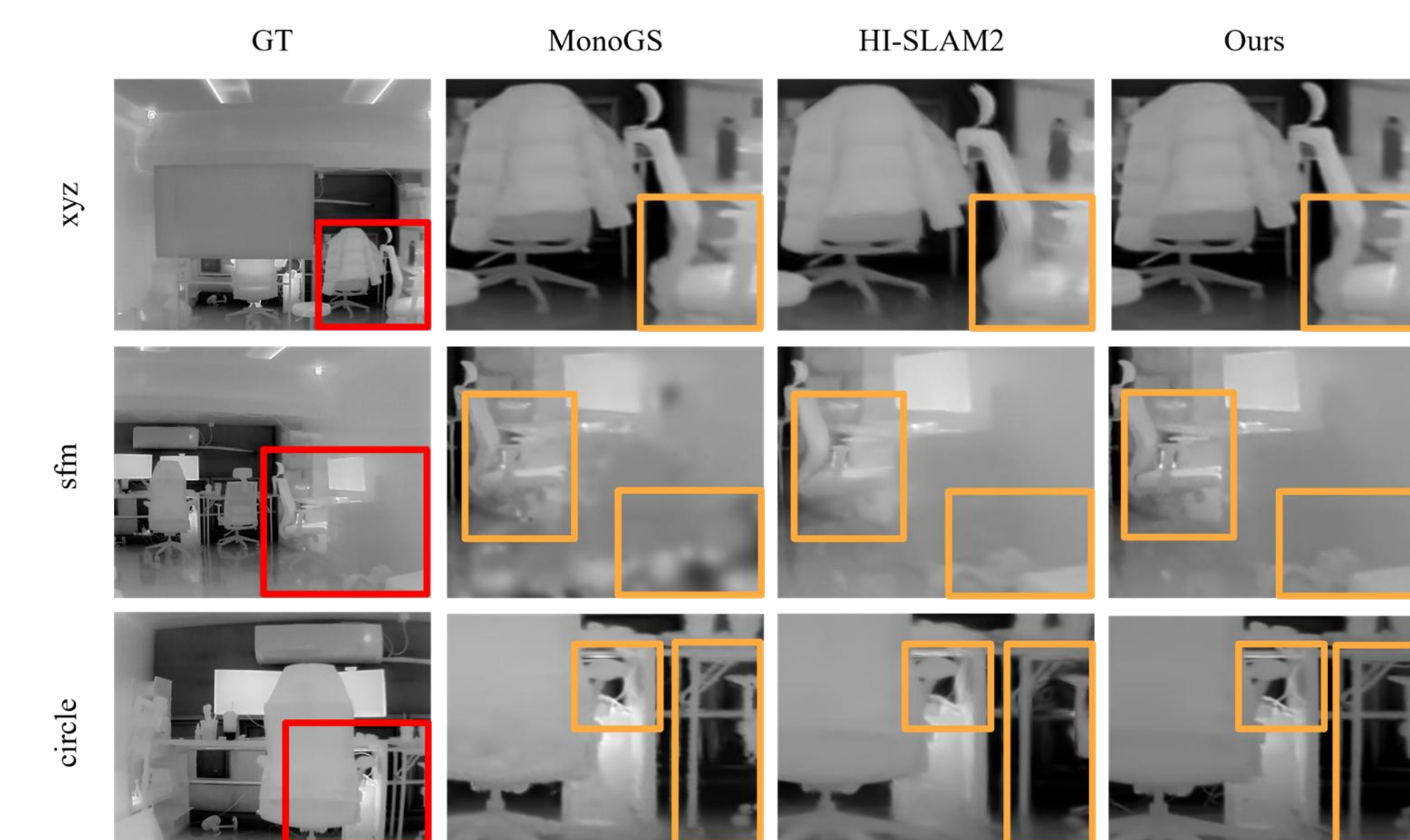
Evaluations

- Multi-spectral dataset

Sequence		ORB2 (RGB-D)	ORB2 (Thermal-D)	MonoGS (Thermal-D)	Thermal-D Odom	Ours
Bright	ds1-xyz	0.012	0.083	0.064	0.072	<u>0.054</u>
	ds1-xyz-ps	0.040	0.141	0.042	0.015	<u>0.031</u>
	ds1-hfsp	0.106	0.442	0.096	<u>0.058</u>	0.049
	ds2-xyz	<u>0.062</u>	Fail	Fail	0.038	0.087
	ds2-hfsp	<u>0.076</u>	0.153	0.154	0.029	0.144
Varying	ds1-xyz-ic	Fail	0.250	<u>0.082</u>	0.115	0.055
	ds1-xyz-ic-ps	Fail	0.349	<u>0.061</u>	0.078	0.052
	ds1-hfsp-ic	Fail	0.467	0.080	<u>0.051</u>	0.045
	ds2-xyz-ic	Fail	0.203	0.171	<u>0.114</u>	0.084
	ds2-hfsp-ic	Fail	0.189	<u>0.099</u>	0.110	0.063



- Custom monocular thermal dataset



- PI-EAT-SLAM demonstrates superior tracking accuracy and competitive NVS performance.

Takeaway Message

- Integrating **thermal physics** and **edge-aware optimization** is key to robust thermal GS-SLAM! 🐱